NEUROMOTION - AI-POWERED REAL-TIME ATHLETE POSTURE ANALYSIS SYSTEM

Project ID: 25-26J-334

Project Final Report

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DECLARATION

I declare that this is my own work, and this thesis does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institution of higher learning, and to the best of my knowledge and belief it does not contain any material previously published or written by another person, except where acknowledgement is made in the text.

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ABSTRACT

Exercise form correction remains challenging for athletes training without continuous professional supervision, as improper movement patterns often develop unnoticed and contribute to preventable injuries over extended training periods. This research introduces NeuroMotion, an intelligent biomechanical analysis platform that leverages two custom-trained YOLOv8 neural networks for real-time exercise classification and 33-keypoint skeletal pose estimation. The system computes joint angles at critical anatomical landmarks using vector-based trigonometry, comparing values against exercise-specific biomechanical thresholds for squats, push-ups, and vertical jumps. Visual guidance is provided via a colour-coded skeleton overlay green for correct alignment, red for detected deviations alongside three horizontal reference lines indicating optimal depth and positioning. Spoken corrective cues are delivered in English, Tamil, and Sinhala via browser-native text-to-speech synthesis, supporting hands-free training for linguistically diverse athletes. Training data comprised over 500 video recordings from more than twenty volunteer participants, partitioned using a 70/15/15 distribution with data augmentation techniques for enhanced model generalisation. Validation against expert annotations yielded 92.1% posture detection accuracy, sub-100-millisecond processing latency on standard processors, and 94.8% repetition counting accuracy. These findings demonstrate that professional-grade biomechanical coaching can be delivered through standard camera hardware alone, eliminating reliance on wearable sensors or motion capture laboratory equipment and making such technology accessible to athletes at all competitive levels.

Keywords: pose estimation, YOLOv8, biomechanical analysis, real-time posture correction, exercise classification, joint angle detection, multilingual voice feedback, sports technology, deep learning

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Special appreciation goes to the more than twenty volunteer participants who generously contributed their time to record exercise footage for the training dataset. Their participation was essential for developing a diverse and representative dataset that enabled the system to generalise across different body types, movement speeds, and exercise variations. Finally, heartfelt thanks are extended to the families and friends of the research team members for their unwavering support, motivation, and understanding during the demanding periods of this project, which helped the team persevere through challenges and maintain commitment to completing this work.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CPU	Central Processing Unit
CSV	Comma-Separated Values
CV	Computer Vision
DL	Deep Learning
EMA	Exponential Moving Average
FPN	Feature Pyramid Network
FPS	Frames Per Second
GIF	Graphics Interchange Format
GPU	Graphics Processing Unit
JSON	JavaScript Object Notation
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
OKS	Object Keypoint Similarity
PDF	Portable Document Format
RGB	Red Green Blue
SLIIT	Sri Lanka Institute of Information Technology
SQL	Structured Query Language
SST	Software Systems and Technologies
TTS	Text-to-Speech
UI	User Interface
UX	User Experience
YOLO	You Only Look Once
CI/CD	Continuous Integration / Continuous Deployment
JWT	JSON Web Token
LiDAR	Light Detection and Ranging
ORM	Object-Relational Mapping
SaaS	Software as a Service
UAT	User Acceptance Testing
CSS	Cascading Style Sheets

1. INTRODUCTION

1.1. Background Study and Literature Review

1.1.1 Background Study

Correct exercise technique has long been understood as central to athletic development, yet the practical means of delivering precise technical feedback to athletes in real time has changed remarkably little over decades. Coaches observe from the sideline; athletes attempt to self-monitor in mirrors or on video replays. Neither approach reliably detects the subtle joint deviations that accumulate into chronic injury across thousands of training repetitions [2], [3].

Traditional coaching methods, while effective when expert trainers are present, suffer from inherent limitations that become particularly pronounced in high-volume training environments. A single coach monitoring multiple athletes cannot simultaneously track the subtle biomechanical deviations that occur during each repetition of a squat, push-up, or jumping movement. These deviations, including knee valgus where the knee collapses inward during squatting movements, excessive forward lean that places undue stress on the lumbar spine, and hip sagging during push-up exercises, often develop into habitual patterns that significantly increase injury risk over time. Studies have consistently demonstrated that improper joint alignment during resistance exercises correlates strongly with increased incidence of patellofemoral pain syndrome, lumbar disc herniation, and shoulder impingement conditions [2], [3].

The emergence of deep learning and computer vision technologies has created unprecedented opportunities to augment traditional coaching with automated, real-time biomechanical analysis systems. Recent advances in pose estimation algorithms, particularly those based on the You Only Look Once (YOLO) architecture, have enabled the detection of human body keypoints with sufficient accuracy and speed to support real-time feedback applications. Unlike earlier generations of motion analysis systems that required expensive marker-based motion capture equipment and controlled laboratory environments, modern deep learning approaches can operate using standard camera input in everyday training settings. This technological progression opens the possibility of bringing professional grade

posture analysis to athletes who previously lacked access to such capabilities due to cost, geographic, or institutional barriers [4], [13], [14].

Transfer learning has emerged as the dominant strategy for adapting pre-trained deep neural networks to domain-specific pose estimation tasks. Rather than training large backbone architectures from random initialisation, researchers have demonstrated that freezing early convolutional layers while fine-tuning higher-level feature representations on labelled domain data produces substantially faster convergence with smaller task-specific datasets [15]. The stacked hourglass architecture proposed by Newell et al. [16] illustrated how hierarchical feature representations, when pre-trained on broad image corpora, could be redirected toward precise keypoint localisation with targeted supervision. This principle informed the two-phase training strategy adopted in NeuroMotion, where the CSPDarknet53 backbone remained frozen for the initial fifty training epochs while the classification and pose heads adapted to exercise-specific features, followed by full network fine-tuning at a reduced learning rate.

Fatigue detection represents an emerging application of motion analysis that extends the utility of exercise tracking systems beyond single-session posture evaluation. Research in this area has demonstrated that measurable changes in movement kinematics reliably precede subjective reports of fatigue: peak joint velocities decline, movement amplitude decreases, and error frequency increases across consecutive repetitions as muscular capacity diminishes [18]. These kinematic signatures are detectable from the same keypoint data used for posture evaluation, suggesting that a unified analysis pipeline can serve both corrective coaching and fatigue monitoring objectives simultaneously. NeuroMotion incorporates a fatigue scoring module within the performance dashboard that tracks declining score trends, increasing bad-move frequency, and reducing movement depth across sessions to flag potential over-training conditions for athlete and coach review.

NeuroMotion represents a research initiative to develop and evaluate an AI-powered real-time athlete posture analysis system specifically designed to address the identified gaps in existing sports technology solutions. The system leverages two custom-trained YOLOv8 models: one dedicated to automatic exercise classification and another for 33-keypoint pose estimation. By computing joint angles through vector trigonometry and evaluating them against exercise-specific biomechanical thresholds, NeuroMotion delivers immediate corrective feedback through both visual skeleton overlays and multilingual voice synthesis. The inclusion of Tamil and Sinhala language support addresses a significant accessibility

gap, as existing fitness technology predominantly serves English-speaking users while neglecting the linguistic diversity of South Asian athletic communities.

1.1.2 Background Survey

Chen et al. examined multi-sensor fusion approaches combining depth cameras and inertial measurement units for human action recognition, demonstrating that sensor fusion improved classification accuracy compared to single-modality sensing in controlled experimental environments [1]. Despite these advances, the fundamental constraint of wearable-based approaches persisted: accelerometers and gyroscopes deliver orientation data but cannot reconstruct three-dimensional skeletal geometry, making them unsuitable for detecting the kind of angular deviations at the knee and hip that constitute meaningful postural error.

Automated exercise analysis draws from four distinct research traditions - wearable sensing, computer vision, deep learning classification, and sports biomechanics - each contributing methods that informed different aspects of the NeuroMotion design pipeline [1], [2], [5], [9].

O'Reilly et al. [2] investigated technology applications in strength and conditioning contexts, specifically examining body weight squat performance assessment using wearable sensors. Their research highlighted the challenge of detecting knee valgus through inertial measurement units alone, as the sensors provide orientation data without direct measurement of joint alignment. The study concluded that while wearables can detect gross movement errors such as insufficient depth or excessive speed, they lack the spatial resolution necessary for identifying the subtle angular deviations that distinguish proper from improper form. This limitation fundamentally constrains the utility of wearable-based systems for posture quality assessment [3].

Markerless pose estimation has advanced substantially since the introduction of OpenPose by Cao et al. [5], which introduced real-time multi-person keypoint detection using part affinity fields. The OpenPose architecture enabled skeleton extraction from monocular video at frame rates sufficient for interactive applications, representing a significant step toward practical real-time coaching systems. However, OpenPose was designed as a general-purpose pose estimation framework without domain-specific optimisation for athletic movements or integrated feedback mechanisms. The system outputs

keypoint coordinates but does not interpret these coordinates in terms of exercise-specific biomechanical correctness.

Bazarevsky et al. [6] subsequently developed BlazePose as a lightweight architecture optimised for on-device inference on mobile devices. The MediaPipe Pose implementation demonstrated that real-time pose tracking could be achieved on consumer hardware without cloud connectivity, addressing the latency and privacy concerns associated with server-based processing. Nevertheless, like OpenPose, BlazePose provides only the raw pose estimation output without exercise classification, posture evaluation, or corrective feedback. The fixed pre-trained weights also limit adaptability to domain-specific movements that may differ from the general activities represented in training datasets.

An analogous domain-specific fine-tuning approach was demonstrated by Liao et al. [4] for physical rehabilitation exercise assessment, where a deep learning framework adapted general pose representations to distinguish correct from incorrect therapeutic movements confirming the viability of domain fine-tuning that NeuroMotion extends to athletic exercise contexts.

Fang et al. [7] proposed RMPE (Regional Multi-person Pose Estimation) to address challenges in multi-person scenarios where occlusions and close proximity cause detection difficulties. Their approach achieved state-of-the-art accuracy on benchmark datasets through symmetric spatial transformer networks and pose-aware reinforcement. While RMPE advanced the technical capabilities of pose estimation systems, it maintained the same limitation as competing architectures: the absence of exercise-specific evaluation logic and feedback generation. Bridgeman et al. [8] extended multi-person pose estimation to three-dimensional tracking for sports applications, demonstrating impressive results in team sports analysis but requiring multiple camera viewpoints unsuitable for individual training contexts.

Exercise classification and recognition have been investigated through various deep learning approaches. Ignatov [9] demonstrated that convolutional neural networks could classify human activities from accelerometer data with high accuracy, though this work focused on activity recognition rather than movement quality assessment. Pestana et al. [10] proposed an automatic exercise classification system using deep learning techniques that achieved strong classification performance on curated datasets. However, their system explicitly acknowledged the inability to distinguish between properly and improperly

executed movements, highlighting a critical gap between knowing what exercise was performed versus whether it was performed correctly.

The YOLO architecture has evolved significantly since its original introduction by Redmon et al. [13], with each iteration improving detection accuracy and inference speed. Wang et al. [11] presented YOLOv7, which introduced trainable architectural modifications that advanced the state-of-the-art for real-time object detection. Jocher et al. [14] subsequently released YOLOv8 through Ultralytics, which for the first time provided native support for both classification and pose estimation tasks within a unified framework. This architectural advancement is particularly significant for the present research, as it enables the development of integrated systems that combine exercise classification with keypoint detection without the complexity of maintaining separate model pipelines. Parallel development of the Single Shot MultiBox Detector [20] contributed complementary insights into the speed-accuracy tradeoffs of one-stage detectors, influencing the architectural refinements incorporated into subsequent YOLO generations.

A notable gap in the existing literature concerns multilingual accessibility in sports technology applications. The vast majority of fitness applications and coaching systems operate exclusively in English, effectively excluding athletes who prefer instruction in their native languages. For South Asian communities, where Tamil and Sinhala represent major linguistic groups, this creates a barrier to adopting technology-assisted training. The integration of multilingual voice synthesis using browser-native text-to-speech capabilities represents an underexplored opportunity to expand the reach of automated coaching systems to linguistically diverse populations. NeuroMotion addresses this gap by implementing comprehensive voice feedback in English, Tamil, and Sinhala, enabling hands-free coaching that accommodates athletes across language preferences.

1.2 Research Gap

AI-driven personal training systems have been reviewed for their capability to personalise workout recommendations, yet none of the surveyed platforms integrated real-time biomechanical posture evaluation during live exercise sessions [18].

Analysis of the existing literature reveals a significant gap between the technical capabilities demonstrated by individual research contributions and the integrated functionality required for practical real-time coaching applications. While substantial

progress has been made in pose estimation accuracy, exercise classification performance, and wearable sensor data analysis, no existing system combines all of the following capabilities: camera-only operation without wearable sensors, real time joint level posture feedback, exercise-specific biomechanical thresholds, and multilingual voice coaching [18], [19].

Table 1. Comparison of Existing Exercise Analysis Systems with NeuroMotion

Feature	Wearable Systems	MediaPipe Apps	Motion Capture Labs	NeuroMotion
Exercise-specific corrective feedback	No	No	No	Yes
Camera-only operation (no wearables)	No	Yes	No	Yes
Joint angle analysis	Partial	No	Yes	Yes
Exercise-specific thresholds	No	No	Yes	Yes
Multilingual voice feedback	No	No	No	Yes
Performance dashboard	No	Partial	Yes	Yes
Affordable for individuals	No	Yes	No	Yes
Trainable/Extensible Models	No	No	Partial	Yes

The research gap identified centres on the absence of an integrated solution that delivers professional-grade biomechanical analysis through readily available hardware. Wearable systems provide some motion data but cannot reconstruct skeletal alignment. Pose estimation libraries output keypoint coordinates without interpreting posture quality. Motion capture laboratories offer accurate analysis but at prohibitive cost and complexity. NeuroMotion fills this gap by combining custom-trained YOLOv8 models for classification and pose estimation, biomechanical threshold evaluation logic, and multilingual voice feedback within a browser-based application accessible to any athlete with a standard camera.

1.3 Research Problem

In the Sri Lankan educational and athletic context, the General Certificate of Education Advanced Level represents a critical juncture in student development, and physical education forms an integral component of holistic development. University athletics programmes and commercial fitness centres across the country face a common challenge: the majority of athletes train without access to real-time, expert-level feedback on their

exercise technique. The consequences of this gap manifest in injury statistics that reveal alarming rates of preventable musculoskeletal conditions among physically active populations.

Sports medicine research has established clear links between specific biomechanical errors and injury outcomes. Knee valgus during squatting movements places excessive stress on the medial collateral ligament and patellofemoral joint, contributing to anterior knee pain and increasing the risk of anterior cruciate ligament injuries. Forward lean exceeding biomechanically safe thresholds during both squats and jumps places undue load on the lumbar spine, accelerating disc degeneration and contributing to chronic lower back pain conditions. Hip sagging during push-up exercises indicates core instability while creating harmful stress patterns in the shoulder complex. Despite the well-documented risks, athletes continue to perform exercises with these errors because they lack the real-time feedback necessary to recognise and correct their form [2], [3], [17].

The coaching access gap exacerbates this problem significantly. Professional sports coaches command fees that place their services beyond the reach of most student athletes and recreational fitness enthusiasts. Geographic distribution of qualified trainers concentrates expertise in urban centres, leaving athletes in rural and suburban areas without access to quality technical instruction. Even when coaches are available, the ratio of athletes to instructors in typical training environments means that individual attention is necessarily limited. A coach overseeing a group training session cannot monitor every repetition of every athlete simultaneously, creating windows during which errors go uncorrected.

Existing digital solutions fail to address this problem adequately. Consumer fitness applications offer workout tracking and basic repetition counting but provide no meaningful assessment of movement quality. An athlete might complete one hundred push-ups according to such an application while maintaining poor form throughout, receiving positive feedback for a session that reinforced harmful movement patterns. Wearable fitness devices track heart rate, steps, and sleep but cannot detect whether a squat was performed with proper knee alignment or whether a push-up maintained a rigid plank position. The technology industry has prioritised activity quantification over movement quality assessment, leaving athletes with sophisticated tracking tools that fail to address the fundamental question of whether they are exercising correctly.

Language barriers present an additional dimension of this problem, particularly relevant in multilingual societies such as Sri Lanka. Athletes who are more comfortable in Tamil or Sinhala than in English face additional obstacles in accessing technology-assisted coaching, as virtually all existing fitness applications operate exclusively in English. This linguistic exclusion limits the adoption of beneficial technology among populations that could benefit significantly from automated coaching assistance. The combination of cost barriers, geographic limitations, and linguistic exclusion creates a multi-dimensional access problem that prevents athletes at all levels from receiving the real-time technical feedback necessary for safe and effective training.

This research addresses the identified problem by developing NeuroMotion, an AI-powered real-time posture analysis system designed to deliver professional-grade biomechanical coaching through accessible technology. The system requires only a standard camera, operates in real time without perceptible delay, provides exercise-specific corrective feedback through both visual and auditory channels, and supports multiple languages to serve linguistically diverse user populations. By combining custom-trained deep learning models with biomechanical evaluation logic and multilingual voice synthesis, NeuroMotion aims to democratise access to quality movement coaching for athletes across the competitive spectrum.

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research is to design and develop NeuroMotion, a real-time AI-powered posture analysis system that utilises custom-trained YOLOv8 models to detect biomechanical errors during physical exercises and delivers immediate corrective feedback through visual overlays and multilingual voice synthesis. The system specifically targets three common exercises squats, push-ups, and vertical jumps and provides coaching support in English, Tamil, and Sinhala to serve the linguistically diverse athletic community of Sri Lanka.

1.4.2 Specific Objectives

1. To train a YOLOv8 classification model that automatically identifies which exercise is being performed from video input, distinguishing between squats, push-ups, jumps, and non-exercise states with accuracy exceeding 90 percent.

2. To train a YOLOv8 pose estimation model that detects 33 body keypoints to construct a complete digital skeleton for each detected athlete, enabling comprehensive biomechanical analysis across all major joints.

3. To implement a joint angle calculation engine using vector-based trigonometry with the atan2 formula, deriving angles at knees, hips, elbows, shoulders, and torso for posture evaluation.

4. To define and validate exercise-specific biomechanical thresholds based on sports medicine literature and empirical testing, establishing criteria for detecting knee valgus, forward lean, knee-over-toes, back arch, head position, arm angle, and hip sag errors.

5. To develop a real-time visual feedback system that overlays a colour-coded skeleton on the video feed and displays three horizontal guideline lines providing spatial reference for movement depth and alignment.

6. To build a multilingual voice feedback module supporting English, Tamil, and Sinhala that delivers spoken corrective instructions when postural errors are detected, using browser-native text-to-speech capabilities.

7. To create a performance tracking dashboard with session history, fatigue detection, AI-generated training recommendations, leaderboards, and PDF report generation for comprehensive athlete progress monitoring.

8. To achieve posture detection accuracy of at least 92 percent validated against expert-labelled ground truth data, with frame processing latency below 100 milliseconds for real-time operation.

1.4.3 Business Objectives

1. To make professional-grade biomechanical posture coaching available to sports academies and fitness centres that currently cannot justify the cost or infrastructure of motion capture laboratories, specifically targeting the Sri Lankan university and commercial gym market.

2. To position NeuroMotion as a scalable software-as-a-service platform for multiple sports academies, with future cloud deployment on Amazon Web Services or Google Cloud Run supporting concurrent multi-user access.

3. To generate sustainable revenue through individual athlete subscriptions, institutional licenses for gyms and university athletic departments, and premium feature offerings including personalised mentorship and advanced analytics dashboards.

4. To expand language support to additional South Asian languages including Hindi, Bengali, and Malayalam for broader market reach across the region, while maintaining the quality of biomechanical analysis.

5. To enable future mobile deployment through model quantization and pruning techniques, reducing YOLOv8 model size below 10 megabytes for on-device inference on consumer smartphones.

2. METHODOLOGY

2.1 Introduction

This chapter presents the comprehensive methodology employed in developing the NeuroMotion real-time athlete posture analysis system. The methodology encompasses system architecture design, the four core components comprising exercise classification, pose estimation with angle calculation, posture evaluation with feedback delivery, and the performance dashboard with AI-powered features. Additionally, the development process following Agile methodology, technical implementation details of model training and deployment, commercialisation considerations, and the testing framework are described in detail.

2.2 System Overview

NeuroMotion operates as a browser-based Python application built with the Streamlit framework, providing a responsive web interface accessible from any modern browser. The system supports three operational modes: single-player live training using a connected camera, video upload analysis for post-session review, and a competitive 1v1 multiplayer mode where two athletes train simultaneously. The backend executes YOLOv8 inference for both exercise classification and pose estimation, with all processing performed on the local machine to ensure real-time responsiveness without network dependency.

The architecture follows a layered design pattern comprising four primary tiers. The user interface layer handles authentication, exercise selection, live detection view, and access to the dashboard, leaderboard, training plans, and AI TrainBot features. The application layer manages session lifecycle, performance prediction using regression models, recommendation generation, report production, and multiplayer coordination. The analysis engine contains the exercise classifiers, pose estimator, angle calculator, posture evaluator, and rendering components for skeleton and guideline visualisation. The feedback layer delivers visual overlays and multilingual voice synthesis, while the data layer persists session records, user profiles, and training recommendations in an SQLite database.

Figure. 1 illustrates the complete system architecture spanning input capture through user interaction layers. Video frames from the camera or uploaded files enter the processing

pipeline at the analysis engine, where the exercise classifier determines the current activity before routing to the appropriate exercise-specific detector. The pose estimator extracts 33 keypoints, which feed into the angle calculator and subsequently the posture evaluator. Visual feedback is rendered directly on the frame, while voice feedback is triggered through browser-based speech synthesis when postural errors are detected.

Figure 1. System Architecture of NeuroMotion

2.3 Component Overview

The NeuroMotion system comprises four primary components, each responsible for a distinct aspect of the posture analysis pipeline. Figure. 2 presents an overview of how these components integrate to deliver the complete functionality.

Figure 2. Component Overview Diagram

2.3.1 Component 1 – Real-Time Exercise Detection and Classification Using YOLOv8

The first component addresses the fundamental requirement of automatically identifying which exercise the athlete is performing from video input. This classification capability is essential because the posture thresholds and evaluation logic differ significantly between exercises a proper squat requires different joint angle ranges than a proper push-up. Without accurate classification, the system cannot apply the appropriate biomechanical criteria.

The component utilises the YOLOv8m-cls architecture, the medium-scale variant selected because it achieves a practical compromise between classification accuracy and the inference speed required for sustained real-time video processing on consumer CPU hardware without a dedicated graphics card [14]. The model accepts input frames at 224×224 resolution, applying global average pooling followed by two fully connected layers of 512 and 128 neurons respectively, with ReLU activation and dropout regularisation at a rate of 0.3. The final softmax layer outputs a four-class probability distribution corresponding to squat, push-up, jump, and none categories.

Transfer learning was employed using ImageNet pre-trained weights, with the backbone frozen for the first 50 epochs while the classification head adapted to the exercise domain. Fine-tuning continued for an additional 50 epochs with a reduced learning rate,

allowing the entire network to specialise for exercise-specific features. Transfer learning from ImageNet pre-trained weights, following the residual connection principles established by He et al. , allowed the classification head to converge reliably within 100 epochs, achieving 90 percent top-1 accuracy on the held-out test set [15].

Table 2. YOLOv8 Classification Model Training Configuration

Parameter	Value
Architecture	YOLOv8m-cls
Input Resolution	224 × 224
Epochs	100
Batch Size	32
Optimizer	SGD with cosine annealing
Initial Learning Rate	0.01
Final Learning Rate	0.001
Transfer Learning	ImageNet pre-trained weights
Classes	squat, push-up, jump, none

2.3.2 Component 2 – YOLOv8 Pose Estimation and Biomechanical Joint Angle Computation

The second component implements the core pose estimation functionality that detects 33 body keypoints to construct a complete digital skeleton of the athlete. This extended keypoint set exceeds the standard 17-point COCO format to capture finer details necessary for comprehensive exercise analysis, including separate landmarks for heels, foot indices, ears, and additional extremity points that improve robustness during floor based exercises.

The pose model architecture employs YOLOv8m-pose, which extends the detection framework to predict keypoint heatmaps alongside object bounding boxes. The CSPDarknet53 backbone extracts features at multiple scales through a Feature Pyramid Network [21] processing at 80×80, 40×40, and 20×20 resolutions. The pose head branches from FPN features through convolutional layers producing a 33×3 output tensor representing x-coordinate, y-coordinate, and confidence for each keypoint [14].

To suppress frame-to-frame detection noise, each keypoint position is smoothed using an Exponential Moving Average (EMA) filter:

$$\hat{P}(t) = \alpha \cdot P(t) + (1 - \alpha) \cdot \hat{P}(t - 1) \quad (1)$$

where $\alpha = 0.7$ is the smoothing factor, $P(t)$ is the raw detected keypoint coordinate at frame t, $\hat{P}(t)$ is the smoothed output, and $\hat{P}(t - 1)$ is the prior smoothed value.

A smoothing factor of 0.7 was determined through iterative user testing, values below this threshold produced visible lag between the athlete's movement and the skeleton response, while higher values failed to filter the positional oscillations arising from frame-to-frame detection inconsistency.

$$\theta_raw = |atan2(C.y - B.y, C.x - B.x) - atan2(A.y - B.y, A.x - B.x)| \quad (2)$$

$$\theta = \min(\theta_raw, 360° - \theta_raw) \quad (3)$$

where A is the proximal keypoint, B is the joint centre, and C is the distal keypoint. Equation (2) computes the unsigned angular difference. Equation (3) normalises to the interior angle, constraining $\theta \in [0°, 180°]$.

Table 3. YOLOv8 Pose Model Training Configuration

Parameter	Value
Architecture	YOLOv8m-pose
Input Resolution	640 × 640
Epochs	150
Batch Size	16
Optimizer	SGD with cosine annealing
Initial Learning Rate	0.01
Final Learning Rate	0.0001
Transfer Learning	COCO-pose pre-trained weights
Keypoints	33 per person
Object Keypoint Similarity	0.80

Key angles computed include the knee angle (hip–knee–ankle) for valgus detection, hip angle (shoulder–hip–knee) for forward lean assessment, elbow angle (shoulder–elbow–wrist) for push-up arm positioning, and torso angle (shoulder–hip–vertical) for back arch detection. Each angle is evaluated against exercise-specific thresholds to determine whether the current posture represents correct form or requires correction. Figure. 3 illustrates the vector geometry of the joint angle calculation, showing the three-keypoint triplet (A–B–C) with vectors v_BA and v_BC, and the resulting interior angle θ at joint B for a representative knee configuration.

Figure 3. Joint Angle Calculation Diagram

2.3.3 Component 3 – Real-Time Posture Evaluation with Visual Overlay and Multilingual Voice Feedback

The third component implements the posture evaluation logic that compares computed joint angles against biomechanical thresholds, triggering visual and voice feedback when violations are detected. The thresholds were established through a combination of sports medicine literature review and empirical testing with experienced athletic trainers [19].

Table 4. Biomechanical Posture Thresholds by Exercise

Exercise	Check	Condition	Threshold
Squat	Knee Valgus	Hip–Knee–Ankle deviation from 180°	> 15°
Squat	Forward Lean	Shoulder–Hip–Knee angle	< 150°
Squat	Knee Over Toes	Knee X − Ankle X	> 10% frame width
Squat	Back Arch	Shoulder–Hip angle	< 160°
Push-up	Back Arch	Same as squat	< 160°
Push-up	Head Position	Nose–Shoulder offset	> 10% frame height
Push-up	Arm Angle	Shoulder–Elbow–Wrist	Outside 90°–150°
Push-up	Hip Sag	Hip below shoulder–ankle line	> 10%
Jump	Knee Valgus	Same as squat	> 15°
Jump	Forward Lean	Same as squat	< 150°
Jump	Knee Over Toes	Same as squat	> 10% frame width

Visual feedback is rendered as a colour-coded skeleton overlay drawn directly on each processed frame. When all posture checks pass, the skeleton connections appear in green, indicating correct form. When one or more violations are detected, the affected joint connections change to red, immediately alerting the athlete to the problem area. Three horizontal guideline lines provide additional spatial reference: a yellow line representing the calibrated reference baseline, a blue line tracking the current movement position, and a green line indicating hip centre alignment.

$$\text{Score_rep} = \max(0, 10 - 2 \cdot \text{N_errors}) \quad (5)$$

where N_errors is the count of distinct posture error types detected during that repetition. The max(0, ·) bound prevents negative per-repetition accumulation.

The voice feedback module utilises the browser's Web Speech API to deliver spoken corrections in the athlete's selected language. A rate-limiting mechanism ensures that alerts are spaced at least four seconds apart, preventing overwhelming feedback during concentrated error periods. The translation dictionary maps each English error message to equivalent Tamil and Sinhala phrases, with language selection persisted across sessions. For example, the LEFT KNEE VALGUS message corresponds to 'இடது முழங்கால் வளைந்துள்ளது' in Tamil and 'වම් දණහිස ඇතුළට නැමී ඇත' in Sinhala.

2.3.4 Component 4 – Performance Tracking Dashboard and AI-Powered Training Features

The fourth component provides comprehensive performance tracking and AI-enhanced features that extend the system beyond real-time feedback into longitudinal athlete development. The performance prediction module employs scikit-learn regression models trained on historical session data to forecast future speed in repetitions per minute, endurance scores, and overall performance ratings.

The AI TrainBot integrates with the Groq API to provide a conversational training assistant capable of answering sport-specific questions and offering personalised advice based on the athlete's performance history [23]. This feature enables athletes to receive guidance on training progression, injury prevention strategies, and technique refinement without requiring immediate human coach availability.

The recommendation engine analyses session performance patterns to generate personalised workout plans and dietary suggestions. When the system identifies consistent errors in specific movements, it produces targeted practice recommendations to address those weaknesses. The training plan module creates structured programmes balancing different exercise types based on the athlete's goals and identified improvement areas.

Additional features include a 3D muscle map visualisation showing which muscle groups are engaged during each exercise, a global leaderboard ranking athletes by exercise type and total points, a 1v1 multiplayer mode enabling competitive training sessions between two simultaneous users, and PDF report generation using FPDF2 to produce session summaries for external review by coaches or sports medicine professionals.

2.4 Development Process

The NeuroMotion development followed Agile methodology, which proved well-suited to the iterative nature of machine learning model development and the requirement for continuous stakeholder feedback. The project was structured into eleven phases spanning approximately 24 weeks, with regular sprint cycles enabling incremental feature delivery and adaptive planning based on emerging technical insights.

Project initiation established the research objectives, scoped the system to three exercises and three languages, and defined the target performance metrics including 92

percent posture detection accuracy and sub-100ms frame processing latency. The product backlog was created containing prioritised features organised into epics for each major component, with user stories detailing acceptance criteria for individual functional elements.

Sprint planning divided the development timeline into focused periods, typically two to four weeks each, with clearly defined deliverables. Daily standup meetings maintained team alignment, facilitated rapid problem resolution, and ensured consistent progress across parallel workstreams in model development, backend infrastructure, and frontend implementation. Collaboration between developers, domain experts in sports biomechanics, and user experience designers proceeded continuously throughout the project lifecycle.

The development and testing phase proceeded in parallel, with continuous integration ensuring code compatibility as features merged into the main branch. Unit tests covered database operations, authentication logic, prediction calculations, recommendation generation, leaderboard queries, and API integrations. Real-time testing validated calibration accuracy, repetition counting against manual tallies, and system stability during extended sessions.

Version control was managed through GitLab, providing repository hosting, merge request workflows, and issue tracking. Microsoft Planner complemented this with visual task management, enabling progress monitoring and deadline tracking across the distributed team. The combination of these tools supported transparent project status visibility and efficient collaboration throughout the development process.

Regular collaboration sessions with supervisors and sports science staff shaped design decisions throughout the project. Fortnightly supervisor reviews provided structured feedback on component progress, and key architectural decisions including the replacement of MediaPipe with custom YOLOv8 models after initial accuracy assessments emerged directly from these consultations. Domain input from sports science staff assisted in validating biomechanical threshold values before implementation, ensuring posture correction criteria reflected established exercise medicine standards rather than arbitrary engineering choices [19].

Each sprint concluded with a formal review demonstrating completed features to the supervisor, followed by a retrospective examining process effectiveness. These retrospectives produced concrete improvements, including the adoption of GitLab merge request approval requirements to prevent untested code from entering the main branch, and

the standardisation of minimum unit test coverage thresholds before any component integration.

A GitLab CI/CD pipeline was configured to run the full unit test suite automatically on every push to the main branch, with pipeline failures blocking merges until all tests passed. Deployment automation through shell scripts re-installed dependencies from requirements.txt and restarted the Streamlit server, ensuring all team members tested against the current integrated build.

As individual components were validated, integration progressively expanded the system from a single-exercise demonstration to the complete three-exercise platform with dashboard, AI features, and multiplayer mode. Each integration milestone was validated against both functional requirements and non-functional performance benchmarks, with failing benchmarks logged as high-priority backlog items for the subsequent sprint.

Threshold calibration, voice translation quality, and interface polish continued through the final sprints based on user acceptance testing with student athletes. Corrections to the Tamil and Sinhala translation dictionaries were incorporated following native speaker review, and the dashboard layout was adjusted after usability testing revealed that fatigue charts were not immediately visible on smaller screen resolutions.

2.5 Development Methodology

The dataset for model training comprised over 500 labelled exercise video clips sourced from multiple origins. The COCO Keypoints dataset contributed 250,000 person instances with 17-keypoint annotations that were extended to the 33-keypoint format required by the pose model. The Kaggle Exercise Dataset provided labelled video clips of common exercises, while custom footage was recorded from more than 20 volunteers at 720p resolution and 30 frames per second using standard webcams [12], [9], [10].

Table 5. Dataset Composition and Augmentation

Aspect	Details
Total Video Clips	500+
Volunteer Participants	20+
Recording Resolution	720p
Frame Rate	30 FPS
Dataset Split	70% training, 15% validation, 15% testing
Horizontal Flips	Applied for bilateral symmetry exercises
Brightness Variation	±30%

Scaling Range	0.8× to 1.2×
Perspective Warping	Simulated diverse camera angles

Data annotation was performed using LabelMe and CVAT tools configured with a 33-keypoint skeleton template. Four exercise classes were defined jump, squat, pushup, and none with the none class containing non-exercise footage to prevent false-positive exercise detection during idle periods. The annotation process required approximately three weeks to complete, with quality verification by multiple annotators to ensure consistency.

The YOLOv8 architecture was customised for the dual-task requirement through a multi-head configuration. The CSPDarknet53 backbone extracts features which are processed by a Feature Pyramid Network at three spatial resolutions. For classification, the network head outputs a 4-dimensional probability distribution through global average pooling followed by fully connected layers. For pose estimation, the detection head predicts 33 keypoints per person, each comprising x-coordinate, y-coordinate, and confidence components.

$$L_total = \lambda_1 \cdot L_cls + \lambda_2 \cdot L_pose + \lambda_3 \cdot L_conf \quad (4)$$

where $\lambda_1 = 0.3$, $\lambda_2 = 0.5$, and $\lambda_3 = 0.2$ are empirically determined weighting coefficients. L_cls applies cross-entropy loss across the four exercise classification labels. L_pose applies L1 distance between predicted and ground truth keypoint (x, y) coordinates. L_conf applies binary cross-entropy for keypoint visibility prediction. The highest weight assigned to L_pose reflects the primacy of keypoint localisation accuracy for biomechanical angle computation in this application.

The calibration process samples approximately 100 frames during which the athlete stands still in a neutral position facing the camera. This calibration phase, requiring roughly 3.3 seconds at 30 frames per second, establishes baseline body positions for movement detection thresholds. Without calibration, the system cannot distinguish intentional exercise movements from normal standing posture, as the absolute keypoint positions vary significantly between individuals based on height, distance from camera, and body proportions.

Movement state machines govern the detection of exercise phases for each supported activity. The JumpDetector tracks states of Standing, Jumping, and Landed, triggered by knee or ankle positions relative to the calibrated threshold. The SquatDetector monitors

Standing, Squatting Down, and Standing Up phases using hip centre position. The PushupDetector tracks Up Position, Pushing Down, and Pushing Up states based on nose Y-coordinate relative to the baseline.

Table 6. Exercise Detector State Machines

Detector	Tracking Point	States
JumpDetector	Ankle/knee Y-coordinate	Standing → Jumping → Landed
SquatDetector	Hip centre Y-coordinate	Standing → Squatting Down → Standing Up
PushupDetector	Nose Y-coordinate	Up Position → Pushing Down → Pushing Up

Each detector operates as a finite state machine that transitions between phases based on the smoothed tracking point crossing calibrated threshold positions. Repetition counts increment only on completing the full transition back to the starting state, preventing partial movements during hesitation or rest from generating false counts.

The backend infrastructure was implemented in Python 3.10, with database operations handled through a custom SQLite wrapper providing ORM-style access. Authentication utilises bcrypt for password hashing with random salts. The performance prediction module employs scikit-learn regression models, while the recommendation engine generates personalised plans based on identified weaknesses. Report generation uses FPDF2 to produce PDF summaries of session performance.

The frontend was developed using Streamlit, providing a single-page reactive application with sidebar navigation. The main content area displays the live video feed with overlaid skeleton and guidelines, while a side panel shows the posture predictor with real-time status updates. Custom CSS implements a premium dark theme with glassmorphism effects. Voice feedback is injected through streamlit.components.v1.html, which executes JavaScript to invoke the Web Speech API.

2.6 Commercialisation Aspects

The global sports technology market presents substantial opportunity for accessible biomechanical analysis solutions. In Sri Lanka specifically, the growing fitness industry encompasses numerous commercial gyms, university athletic programmes, and sports academies that currently lack access to professional-grade motion analysis tools. NeuroMotion addresses this market gap by providing comparable functionality to laboratory motion capture systems at a fraction of the cost and complexity.

The business model employs a tiered subscription approach designed to maximise accessibility while ensuring sustainable revenue generation. Individual athletes may access basic functionality including repetition counting and performance tracking through a free tier, while premium features such as detailed posture analysis, multilingual voice coaching, and AI-powered recommendations require a monthly or annual subscription. Institutional licenses enable gyms, universities, and sports academies to provide NeuroMotion access to all members at a negotiated rate.

The scalability roadmap progresses through three phases. Initially, deployment occurs on local hardware within individual facilities, with all processing performed on the client machine. The second phase targets cloud deployment on AWS or Google Cloud Run, enabling centralised model serving and database management for multiple concurrent users. Mobile deployment will employ quantization aware training and weight pruning, following lightweight architecture principles demonstrated by MobileNetV2 , targeting a model size below 10MB for on-device smartphone inference [22].

Revenue streams diversify across individual subscriptions, institutional licensing, and premium feature offerings. Individual users may upgrade to access advanced analytics dashboards, personalised mentorship features, and extended session history. Institutional customers receive volume pricing, administrative management tools, and white-label deployment options. Strategic partnerships with sports equipment retailers and fitness certification bodies present additional revenue opportunities through referral programmes and co-branded implementations.

The competitive advantage of NeuroMotion derives from its unique combination of capabilities: camera-only operation eliminating hardware costs, real-time joint-level feedback exceeding what consumer wearables provide, exercise-specific biomechanical thresholds validated against sports medicine research, and multilingual voice coaching serving underserved linguistic communities. No existing solution delivers this complete feature set, positioning NeuroMotion to capture market share in both the Sri Lankan domestic market and broader South Asian region.

Table 7. Technology Stack

Layer	Technology
Backend Language	Python 3.10+
Frontend Framework	Streamlit
AI / Deep Learning	Ultralytics YOLOv8

Computer Vision	OpenCV
Numerical Computing	NumPy
Machine Learning	scikit-learn
Data Visualization	Plotly, Pandas
Database	SQLite
Authentication	bcrypt
Voice Synthesis	Web Speech API (browser-native)
AI Chatbot	Groq API (LLM)
Report Generation	FPDF2

2.7 Testing & Implementation

The testing framework followed a comprehensive pyramid structure progressing from unit tests through integration tests to full system validation. This hierarchical approach ensured that individual components functioned correctly before integration, and that integrated systems behaved as expected before deployment.

Unit testing covered all discrete modules including database operations (test_database.py), authentication logic (test_auth_logic.py), performance prediction calculations (test_prediction.py), recommendation generation (test_recommendations.py), leaderboard queries (test_leaderboard_fix.py), voice alert generation (test_audio.py), and AI chatbot integration (test_llama_integration.py). Tests were executed using pytest with coverage reporting, ensuring that core functionality was validated against expected outputs for representative inputs.

Module testing validated each exercise detector independently. Calibration accuracy tests verified that the 100-frame sampling period established correct baselines for movement thresholds. Known posture errors were deliberately introduced during test sessions to confirm that the system detected them appropriately. Repetition counting was validated against manual tallies by independent human counters, achieving 94.8 percent agreement across 200 test sessions.

Integration testing examined the end-to-end frame processing pipeline from camera capture through classification, pose estimation, angle calculation, posture evaluation, visual rendering, voice synthesis, and database persistence. The 1v1 multiplayer mode was tested with two simultaneous users to verify independent tracking and accurate scoring. Session data integrity was verified by comparing database records against expected values derived from test session inputs.

System testing addressed real-time performance under operational conditions. Calibration accuracy was validated across different body types and camera distances. Repetition counting accuracy was measured against three independent human counters, with inter-annotator agreement of 97.2 percent providing ground truth for comparison. Latency testing measured frame processing time across 1000 consecutive frames, confirming median latency below 100 milliseconds. Extended session testing ran the system continuously for 30 minutes to verify stability, memory management, and absence of crashes or database corruption.

Browser compatibility testing verified voice synthesis functionality across Chrome, Firefox, Safari, and Edge on both desktop and mobile platforms. Tamil and Sinhala voice availability was confirmed on systems with appropriate language packs installed, with fallback messaging provided where native voices were unavailable. User acceptance testing with volunteer athletes assessed task completion rates, system usability scores, and feedback comprehension quality.

3. RESULTS AND DISCUSSION

This chapter presents the experimental results obtained from evaluating the NeuroMotion system across multiple performance dimensions. The evaluation methodology employed a held-out test set of 150 video clips containing multiple exercise repetitions performed by subjects not represented in the training data, ensuring unbiased assessment of generalisation capability.

Table 8. Performance Evaluation Results

Metric	Target	Achieved
Posture Detection Accuracy	≥ 92%	92.1% (150 test clips)
Exercise Classification Accuracy	≥ 90%	90%
Repetition Counting Accuracy	≥ 93%	94.8% (200 rep sessions)
Frame Processing Latency (CPU)	< 100ms	< 100ms
Voice Alert Delivery Rate	≥ 98%	≥ 98%
Calibration Time (100 frames)	≤ 3.3 s	3.3 s
System Uptime (30 min session)	No crashes	Stable

At 30 frames per second, this processing window means corrective feedback reaches the athlete within roughly three frame cycles a short enough interval that the skeleton updates feel immediate during live training.

Repetition counting accuracy of 94.8 percent was validated through comparison with three independent human counters who manually tallied repetitions from the same video sessions. The inter-annotator agreement among human counters reached 97.2 percent, establishing a reliable ground truth benchmark. The system's counting accuracy approaches this human-level performance while providing the additional benefit of simultaneous posture quality assessment that human observers cannot reliably perform at equivalent speed.

Table 9. Posture Detection Accuracy by Exercise Type

Exercise	Checks Performed	Detection Rate
Squat	Knee valgus, forward lean, knee over toes, back arch	92.3%
Push-up	Back arch, head position, arm angle, hip sag	91.1%
Jump	Knee valgus, forward lean, knee over toes	92.8%

The briefly airborne interval, during which neither foot maintains ground contact, reduces lower-limb occlusion and dampens postural sway in the upper body simultaneously, producing a detection signal of substantially higher fidelity for jump alignment evaluation.

As the push-up descends toward its lowest position, the chest, wrists, and elbows converge toward the floor simultaneously, generating a detection challenge in which arm landmarks compete with background for visibility and confidence scores drop for the affected keypoints.

Figure. 4 presents sample output from a squat session showing the skeleton overlay rendered on the video frame. The green skeleton indicates correct posture, while red connections would signal detected errors. The three horizontal guideline lines provide spatial reference for movement depth and alignment.

Figure 4. Squat Posture Correction Output

Figure. 5 demonstrates push-up detection with the nose Y-coordinate serving as the primary tracking point. The performance ribbon at the bottom displays real-time statistics including repetition count, points, and bad moves.

Figure 5. Push-Up Posture Correction Output

Figure. 6 illustrates jump detection with the trigger point (knees or ankles depending on height setting) tracked against the calibrated threshold. The system counts a complete jump when the trigger point rises above and returns below the threshold.

Figure 6. Jump Detection Output

The experimental results confirm that NeuroMotion occupies a distinct position in the existing solution landscape. The 92.1% posture detection accuracy achieved using only standard camera input matches or exceeds what wearable-based systems report for comparable exercises, while the sub-100ms latency demonstrates that the additional computational overhead of joint angle evaluation does not prevent real-time operation [19]. The multilingual voice module performed with a delivery rate above 98% across test sessions, confirming that browser-native speech synthesis provides reliable enough throughput for hands-free training use in all three supported languages.

Figure. 7 displays the performance dashboard presenting historical session data through interactive charts. The dashboard includes daily trend analysis, exercise distribution

breakdown, performance rating trajectories, and fatigue score calculations based on declining performance patterns.

Figure 7. Dashboard Performance Charts

Figure. 8 illustrates the multilingual voice feedback pipeline, showing how detected posture errors are translated into the selected language and delivered through browser-native speech synthesis with rate limiting to prevent alert fatigue.

Figure 8. Voice Alert Translation Flow

Several limitations warrant acknowledgement for future development consideration. The system operates with monocular camera input only, performing 2D analysis without depth estimation, which may reduce accuracy for movements with significant forward-backward motion relative to the camera plane. The SQLite database architecture employs single-writer semantics, making it unsuitable for concurrent multi-user deployment without migration to a server-based database system. Voice synthesis for Tamil and Sinhala depends on browser support, and some platforms may lack native voices for these languages, requiring users to install additional language packs.

3.1 Dashboard and AI Feature Performance

The performance dashboard rendered interactive Plotly charts within one second of page load across all tested configurations, displaying daily performance trends, exercise distribution breakdowns, and fatigue scores derived from declining score patterns across sessions. No data loss was observed during the 30-minute extended session stability test. The AI TrainBot, integrated through the Groq large language model API, returned contextually appropriate responses to exercise-specific and training management queries within approximately two to three seconds per request [23]. The personalised recommendation engine generated structured workout and dietary plans based on identified performance weaknesses, with user acceptance testing participants describing the suggestions as specific and immediately applicable to their training programmes. PDF session reports were generated successfully across all test sessions using the FPDF2 library, producing multi-page summaries of repetition counts, accumulated scores, bad-move breakdowns, and trend visualisations suitable for review by coaching staff.

4. SUMMARY OF EACH STUDENT'S CONTRIBUTION

The NeuroMotion project was developed collaboratively by a team of four students, each taking primary responsibility for specific components while contributing to overall system integration and testing.

Table 10. Summary of Student Contributions

Student Name	Student ID	Component	Primary Role
Manulakshan Yogarajah	IT22283962	Data Pipeline Integration And Pose-Based Jump Analysis	Project Manager / ML Engineer
Mathumitha Gnanasekaran	IT21379338	Computer Vision for Athlete Form & Technique Analysis and Voice Feedback	AI Engineer / Backend Developer
Leashan Marino Sylvester	IT22311368	Design and train AI Model for performance prediction	Computer Vision Engineer / Frontend Developer
Giritharan Jude Jawakker	IT22330864	Build a personalized, intelligent training and visualization system	Full-Stack Developer / Business Analyst

Manulakshan Yogarajah led the design and training of the YOLOv8 exercise classification model, curated and annotated the exercise dataset, managed the Agile sprint schedule, and coordinated version control across the team. His responsibilities included ensuring model convergence, evaluating classification accuracy, and integrating the classification output with the exercise-specific routing logic.

Mathumitha Gnanasekaran developed the YOLOv8 pose estimation pipeline, implemented the joint angle calculation engine using vector trigonometry, and designed the database schema for session persistence. Her work focused on ensuring keypoint detection reliability across varied body types and implementing the Exponential Moving Average smoothing algorithm.

Leashan Marino Sylvester designed the posture evaluation logic, implemented the visual feedback rendering including skeleton overlay and guideline lines, and developed the multilingual voice feedback system. His contributions also included frontend development of the Streamlit interface and browser compatibility testing for voice synthesis.

Giritharan Jude Jawakker implemented the performance dashboard, AI TrainBot integration with Groq API, recommendation engine, PDF report generation, and 1v1

multiplayer mode. He also conducted market analysis and developed the commercialisation strategy for the system.

5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

NeuroMotion demonstrates that real-time biomechanical exercise correction can be achieved using custom-trained YOLOv8 models operating on standard camera input without wearable sensors or specialised laboratory equipment. The system achieves 92.1 percent posture detection accuracy across three common exercises while maintaining sub-100ms frame processing latency suitable for real-time coaching applications. Joint angle computation through vector trigonometry enables exercise-specific threshold evaluation for knee valgus, forward lean, and other posture parameters that directly influence injury risk and training effectiveness.

The specific design decisions underlying these results merit consideration. Custom-trained YOLOv8 models were selected over fixed-weight alternatives such as MediaPipe because domain-specific re-training on exercise footage produced higher accuracy for the narrow movement vocabulary relevant to this application [14]. The exponential moving average smoothing factor of 0.7 was arrived at through iterative testing that balanced responsiveness to genuine movement against suppression of frame-to-frame detection jitter. The two-phase training strategy freezing the backbone initially before full fine-tuning preserved general visual feature extraction while enabling specialisation for exercise-specific keypoint configurations, contributing to the model's ability to generalise across the body type diversity present in the test set [15].

The system makes professional-grade posture coaching accessible to athletes at all competitive levels by eliminating the hardware requirements and costs associated with motion capture laboratories. The 94.8 percent repetition counting accuracy approaches human-level performance while providing the additional capability of simultaneous posture quality assessment. Multilingual voice feedback in English, Tamil, and Sinhala extends accessibility to athletes who prefer hands-free coaching in their native language, addressing a gap in existing fitness technology that predominantly serves English-speaking users.

The research contributes to the sports technology domain by providing an integrated solution combining camera only operation, real-time joint level feedback, exercise specific biomechanical thresholds, and multilingual voice coaching within a single accessible

platform. The modular architecture supports future expansion to additional exercises and languages, while the documented training methodology enables replication and extension by subsequent researchers [18], [19].

The findings suggest that deep learning based pose estimation has matured to the point where professional grade biomechanical analysis can be delivered through consumer hardware, democratising access to movement quality assessment that was previously available only in specialised laboratory settings. This capability has implications for injury prevention programmes, remote coaching applications, and the broader integration of artificial intelligence into athletic training workflows.

For Sri Lanka's multilingual athletic communities specifically, the Tamil and Sinhala voice integration demonstrates that AI coaching technology need not be restricted to English speaking populations a design principle with implications for fitness technology development across the broader South and South-East Asian region.

5.2 Future Work

Future development will focus on cloud-based software-as-a-service deployment targeting multiple sports academies simultaneously. Containerised deployment on AWS or Google Cloud Run will enable sub-50ms inference latency through GPU accelerated processing, supporting concurrent users without the performance limitations of local central processing unit execution. This architectural transition will also enable centralised model updates and feature deployments without requiring individual user action.

A lighter mobile application version is planned for individual athletes and personal trainers, using model quantization and pruning techniques to reduce YOLOv8 model size below 10MB for on device inference. Mobile deployment will extend NeuroMotion's reach to training environments where laptop or desktop computers are impractical, while on-device processing maintains privacy by eliminating the need to transmit video data to external servers [22].

Depth estimation will be incorporated through stereo camera or LiDAR sensing, where available, enabling full three dimensional joint angle computation. This enhancement addresses the current limitation of monocular 2D-only analysis, which may reduce accuracy for movements with significant forward-backward displacement relative to the camera plane.

Depth aware analysis will particularly benefit exercises involving lateral movements and complex multi planar motions.

The exercise library will be expanded beyond the current three supported movements to include additional resistance training, flexibility, and sport specific drills. Priority exercises identified through coach interviews include lunges, deadlifts, overhead press movements, and sport specific drills for cricket and rugby. Each new exercise will require threshold derivation through sports medicine literature review and empirical validation with expert trainers.

Database architecture migration from SQLite to PostgreSQL will support concurrent multi-user deployment for team training environments. This infrastructure change will enable a single NeuroMotion instance to serve entire sports academies or university athletic departments, with role based access controls and administrative dashboards for coaching staff. Additional language support for Hindi, Bengali, and Malayalam will expand market reach across the South Asian region.

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